atomcamp

End-to-End Machine Learning Application

1.0 Project Objective

The objective of this project is to develop an end-to-end machine learning system with an interactive user interface for real-time predictions. The system will leverage the power of GitHub for version control and collaboration, Python for machine learning model development, Streamlit for creating an intuitive web interface, and Streamlit Community Cloud for easy deployment. The project aims to achieve the following:

1.1 Version Control Setup

The first step will involve creating a GitHub repository to manage and maintain version control throughout the project. This repository will serve as the central location for code collaboration, change tracking, and project documentation.

1.2 Local Environment Setup

The project will be developed in a local Python environment using VS Code as the integrated development environment (IDE). A virtual environment will be set up to manage dependencies and avoid conflicts with other Python projects. Essential libraries such as Numpy, Jupyter, Pandas, Scikit-learn, and Streamlit will be installed to support model development and web application creation.

1.3 Model Development

The core of the project will involve building a machine learning model capable of making predictions based on input data. This process will be carried out in a Jupyter Notebook to facilitate interactive data exploration, preprocessing, and model evaluation. The model will be trained on a dataset, validated for accuracy, and optimized through hyperparameter tuning to enhance performance.

1.4 Streamlit UI Development

Following model development, an interactive web interface will be built using Streamlit. The interface will allow users to input their data, trigger predictions from the trained model, and view the results in real-time. This will make the model accessible and usable by non-technical users.

1.5 Deployment

Once the model and interface are developed, the project will be deployed using Streamlit Community Cloud, ensuring that the application is hosted online and easily accessible. The deployment process will involve generating a requirements.txt file to define the necessary Python libraries and versions, ensuring the application can run on the cloud without issues.

1.6 Outcome

The outcome of this project will be a fully functional machine learning application where:

- Users can interact with the model via a simple web interface built with Streamlit.
- The model can make real-time predictions on new data.

- The project will be easily reproducible and maintainable through GitHub version control and cloud-based deployment.
- By the end of this project, participants will have the skills to integrate machine learning models into a fully functional web application. They will gain practical experience in developing an end-to-end system, from data preprocessing and model training to deployment. Participants will also learn to ensure the system's scalability, maintainability, and modularity, while acquiring proficiency in version control with GitHub, environment management, and cloud-based deployment on Streamlit Community Cloud.

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Dataset Card

- **Title:** Cardiotocography (CTG) Dataset — Fetal State Classification
- **Source:** UCI Machine Learning Repository
- **Dataset URL:** https://archive.ics.uci.edu/dataset/193/cardiotocography
- **Creators:** D. Campos, J. Bernardes
- **License:** Creative Commons Attribution 4.0 (CC BY 4.0)
- **Format:** Excel (CTG.xlsx), can be converted to CSV for ML tasks
- **Instances:** 2,126
- **Features:** 21 numeric features derived from fetal heart rate (FHR) and uterine contraction (UC) signals
- **Target:** NSP — Fetal state classification (Normal, Suspect, Pathologic)

Overview

Cardiotocography (CTG) is a standard prenatal monitoring technique that records fetal heart rate (FHR) and uterine contractions (UC) to assess fetal well-being during pregnancy. Detecting abnormal fetal states early is crucial for reducing risks of complications during labor and delivery.

The CTG dataset contains 21 quantitative features extracted from FHR and UC signals collected from 2,126 instances. Each instance is labeled by medical experts with the fetal state (**NSP**): Normal, Suspect, or Pathologic. This makes the dataset ideal for supervised multi-class classification tasks, where the goal is to predict fetal health based on CTG signal features.

Input Features

The dataset includes the following **numeric features**:

Feature	Description	Feature	Description
LB	FHR baseline (beats per minute)	Max	Maximum FHR
AC	Number of accelerations per second	Nmax	Number of histogram peaks
FM	Number of fetal movements per second	Nzeros	Number of zeros in histogram
UC	Number of uterine contractions per second	Mode	Mode of FHR histogram
ASTV	% of abnormal short-term variability	Mean	Mean FHR
MSTV	Mean short-term variability	Median	Median FHR
ALTV	% of abnormal long-term variability	Variance	Variance of FHR
MLTV	Mean long-term variability	Tendency	Trend of FHR
Width	Width of FHR histogram	Mean-value	Average FHR
Min	Minimum FHR	Other histogram/statistics	Additional numeric descriptors of FHR distribution

Target Label

Target	Values	Description
NSP	1, 2, 3	Fetal State: 1 = Normal, 2 = Suspect, 3 = Pathologic

This is a **3-class multi-class classification problem**.

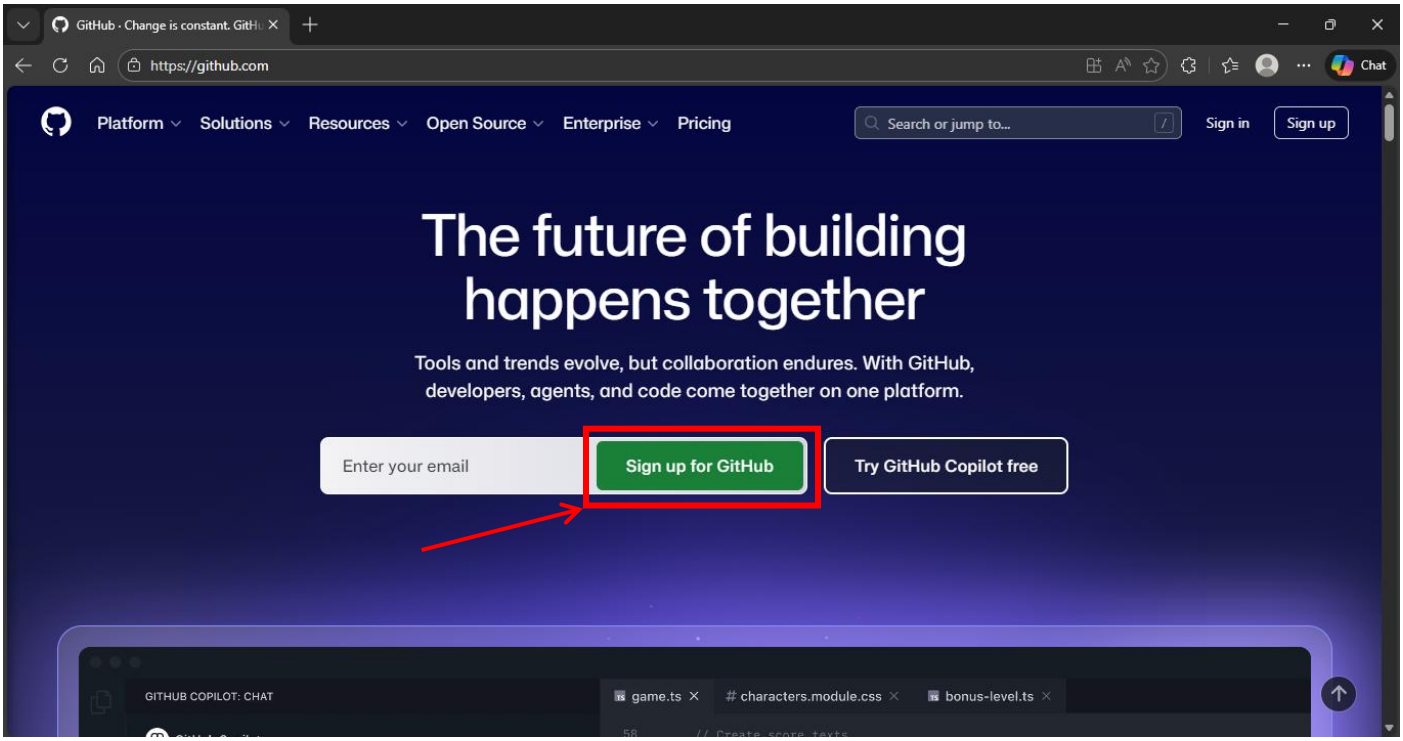
1. Version Control Setup

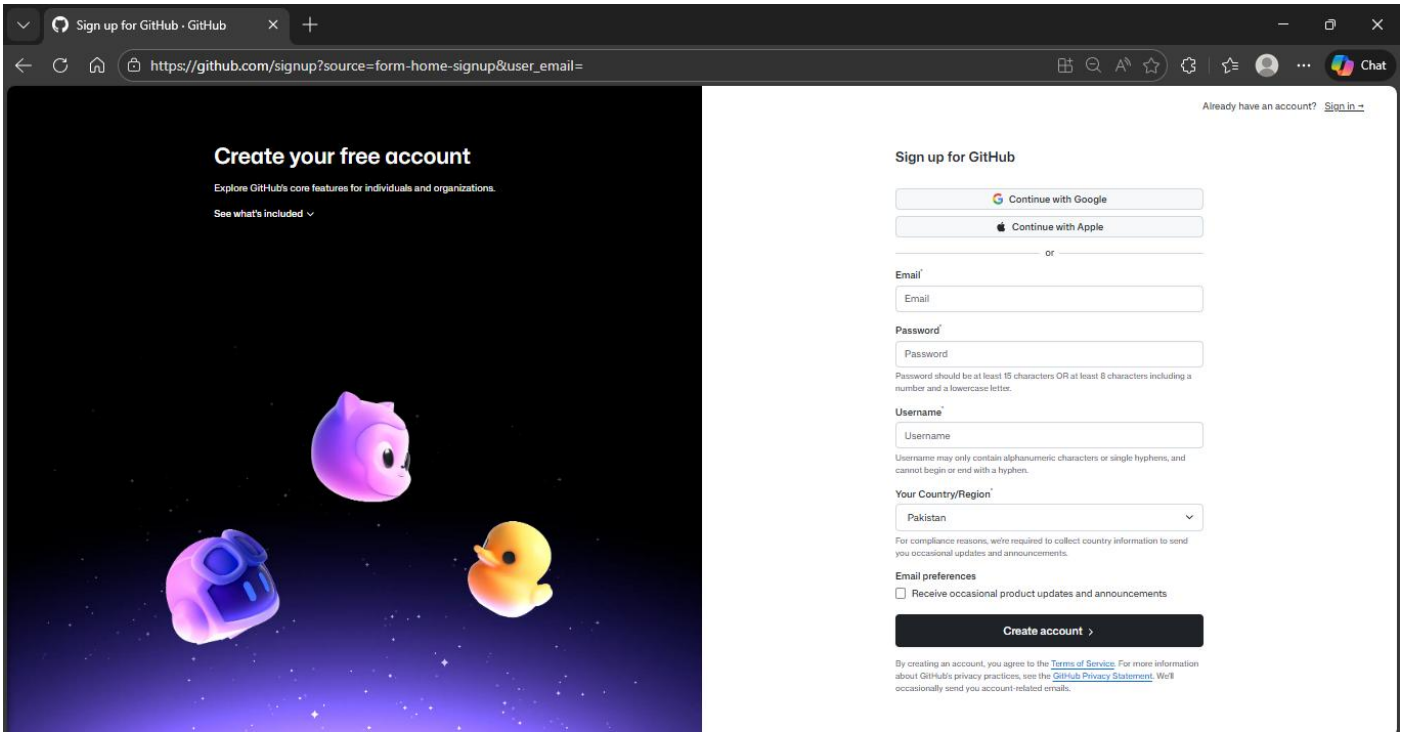
Step 1. Version Control Setup

1.1 GitHub Account Creation

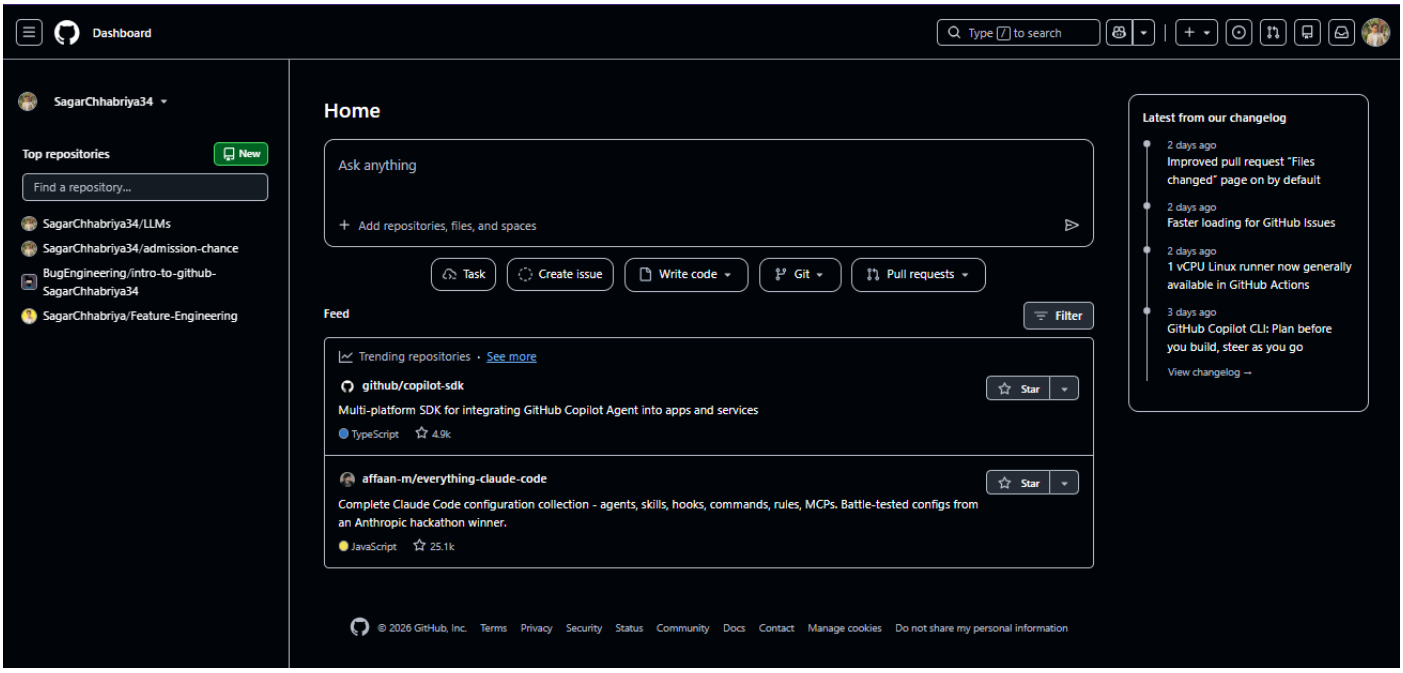
To manage and maintain version control for the project, GitHub will be used as the hosting platform. Follow the steps below to create and configure a GitHub repository:

1. Navigate to https://github.com.
2. Create a new GitHub account using your institutional email address (e.g., .edu) if available; alternatively, a Gmail address is also acceptable.





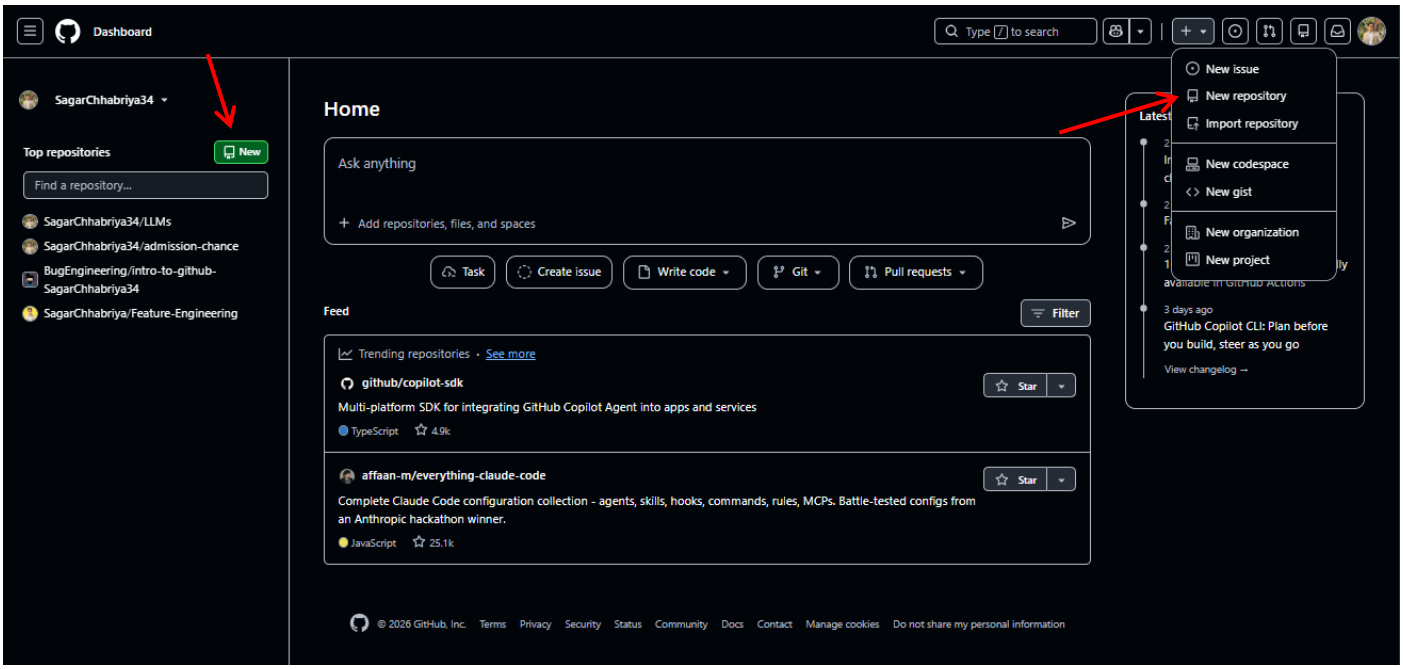
Once registered and logged in, the GitHub home screen will be displayed. If this is your first-time using GitHub, the interface may appear minimal, as no repositories have been created yet.



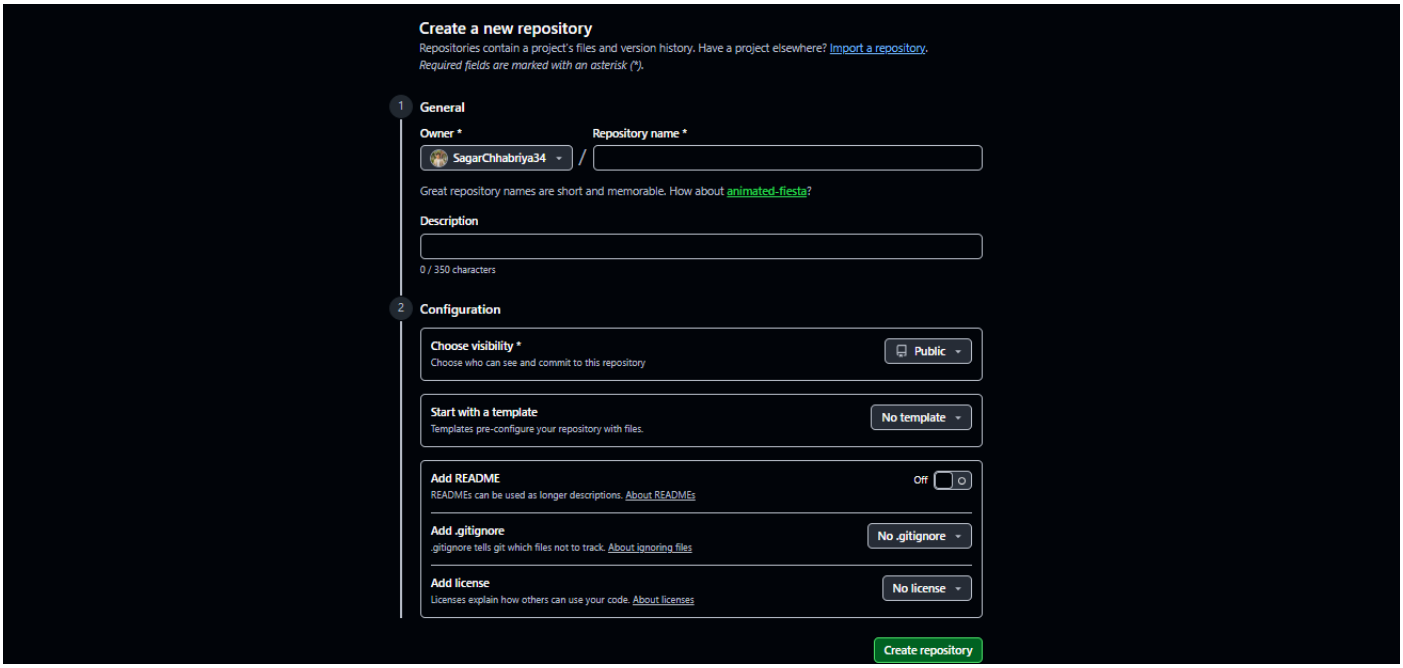
1.2 Creating a New Repository

To create a new repository for the project:

1. On the GitHub homepage, click the green *New* button **or** click on the "+" icon in the top-right corner and select "New repository" from the dropdown.

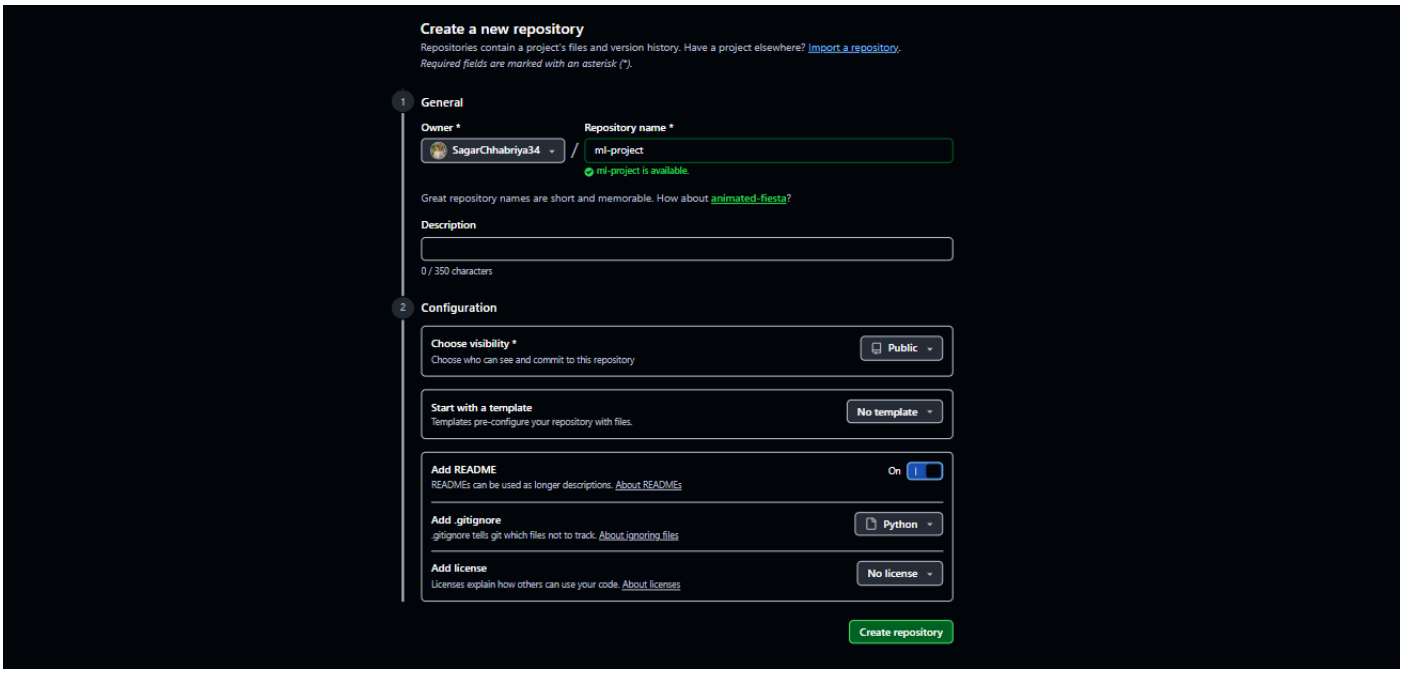


2. Fill in the repository details as follows:

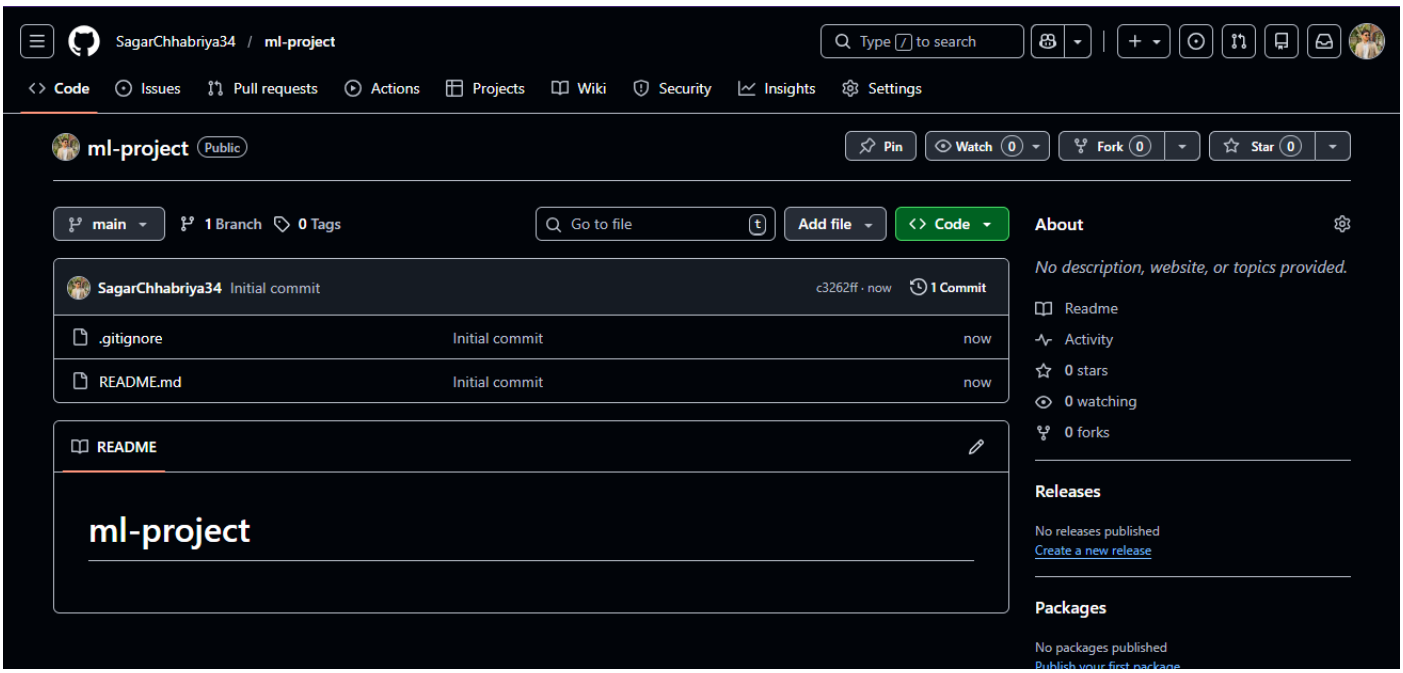


- Repository Name: ml-project
- Description: *(Optional; can be left blank)*
- Visibility: Select Public
- Template: Leave at default (no template)
- Initialize Repository:
  - Add a README file: Enable this option
  - Add .gitignore:
    - Click on the .gitignore dropdown
    - Type Python and select it
  - License: Leave at default (No license selected)

3. Click the green “Create repository” button to complete the setup.



Once the repository is successfully created, it will be initialized with a README.md and a Python-specific .gitignore file, making it ready for cloning and further development.



In the next section, we will install VS Code, an IDE, to set up our local environment.

2. Dev Setup (VS Code)

Step 2. Local Setup (VS Code)

To set up the local development environment for this project, follow the steps below:

2.1 Install Required Software

1. **Python Installation**:

   Ensure that Python is installed on your local system. If not, download and install the latest stable version from Python's official website.

   - During installation, ensure to check the option **"Add Python to PATH"** to set the necessary environment variables automatically. How to install Python and VS Code?

2. **Install Visual Studio Code (VS Code)**:

   Download and install VS Code, which will be used as the primary integrated development environment (IDE) for this project.

3. **Install Required Extensions for VS Code**:

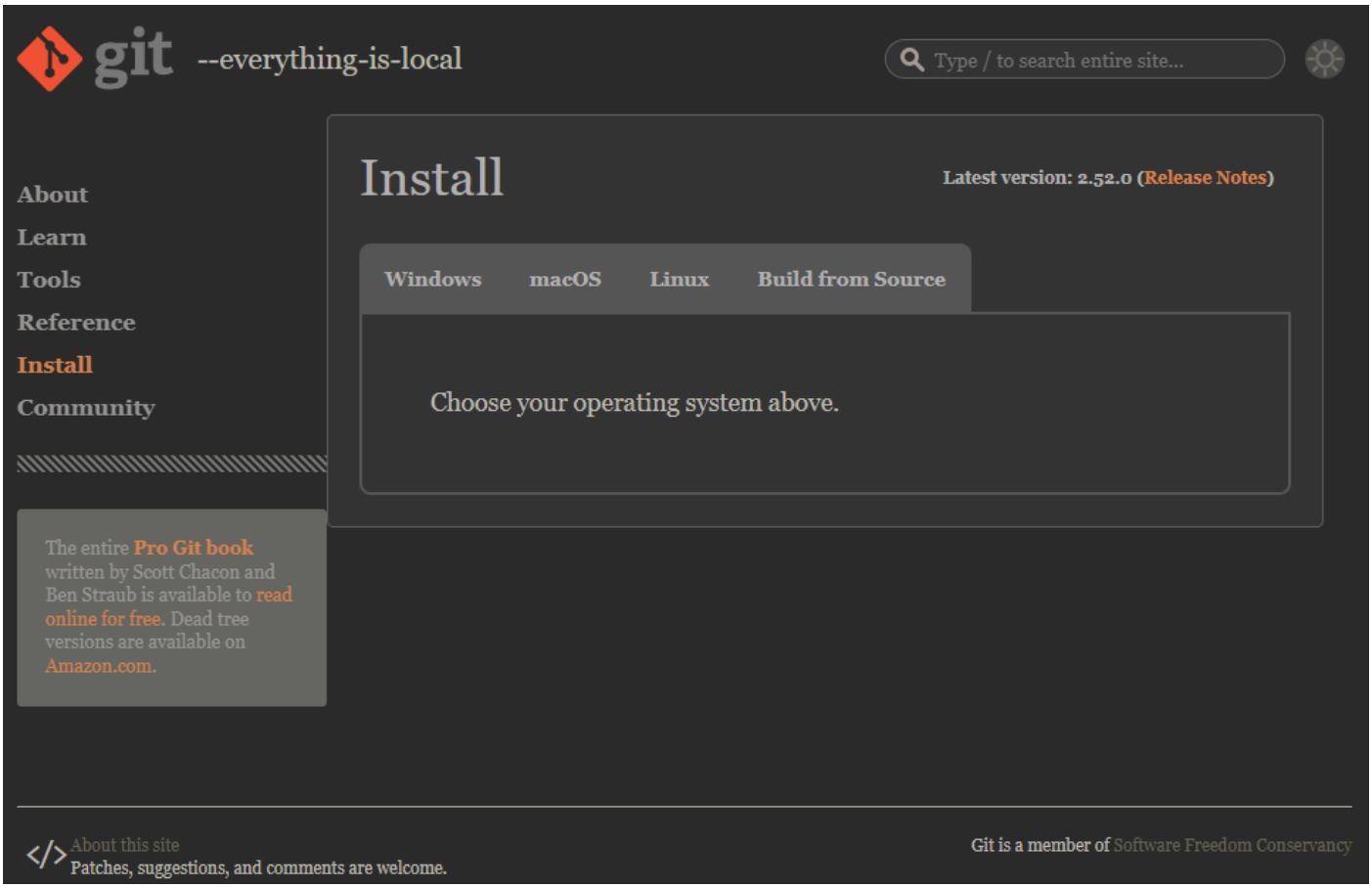
   - Open VS Code and go to the **Extensions** tab on the left sidebar.
   - Install the following extensions:
     - **Python** (for Python development)
     - **Jupyter** (for working with Jupyter notebooks within VS Code)

In this case, it's already installed, and it shows the disable option.

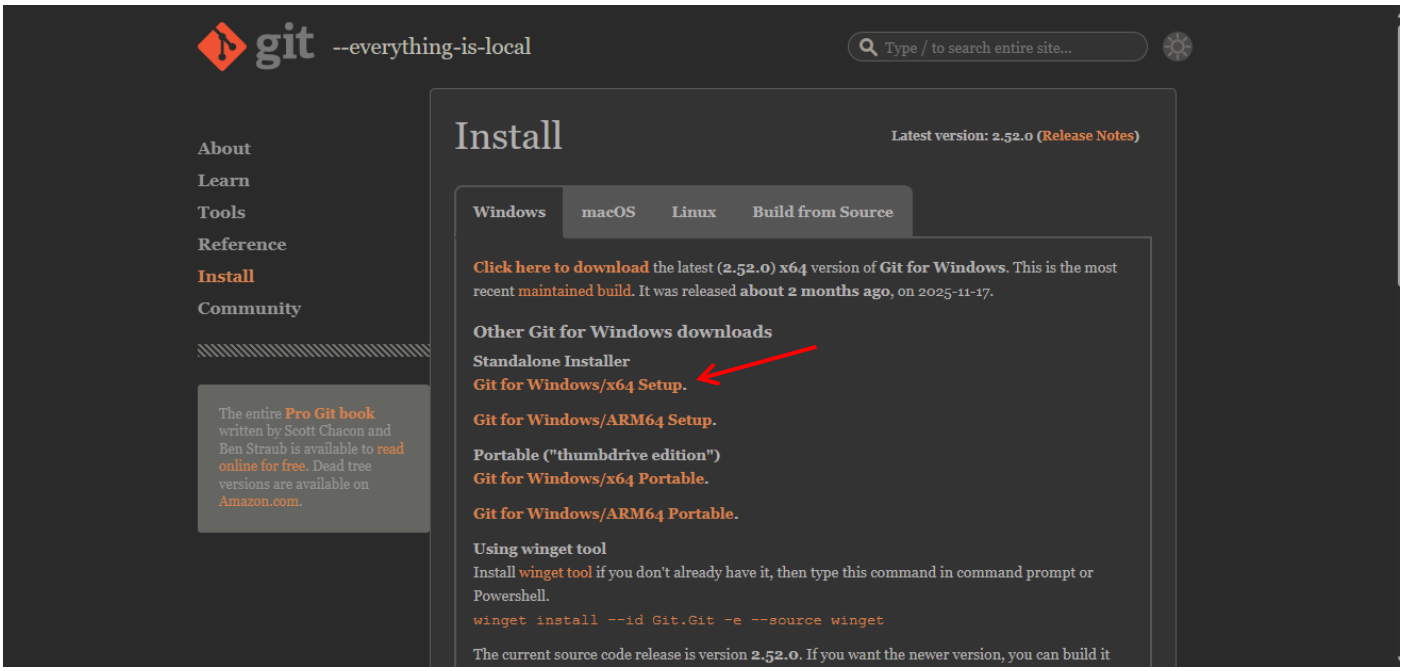
4. **Download and install Git Bash** (if not already installed):

   Git Bash is a terminal emulator that allows you to run Git commands and other shell commands on Windows.

Follow these steps to install it: Go to the official Git Downloads page.

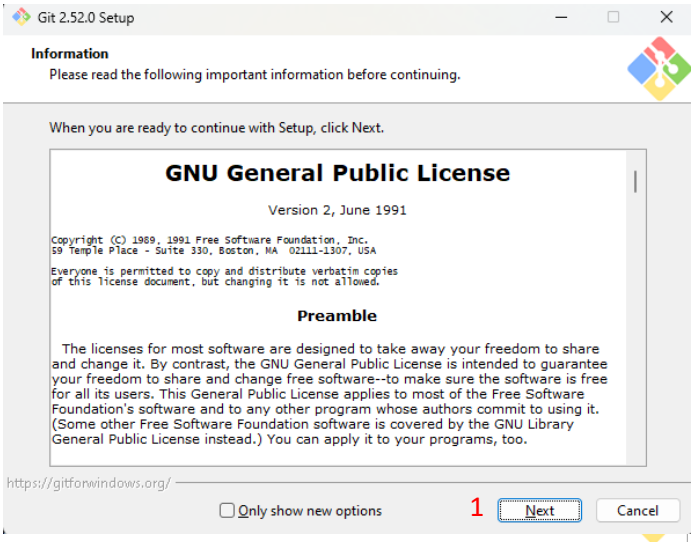


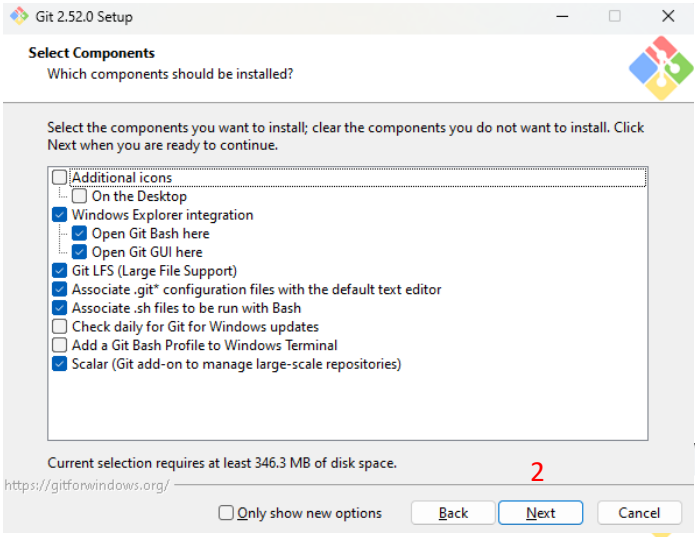
Select **Windows** and download the latest version of Git for Windows.

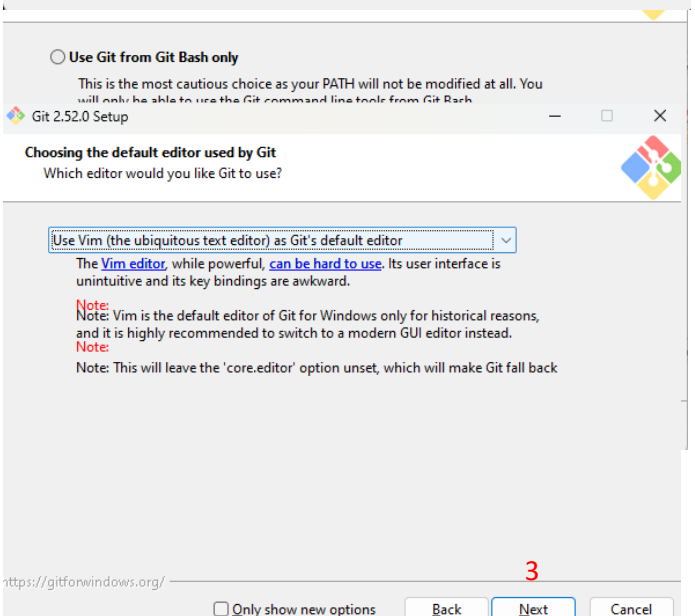


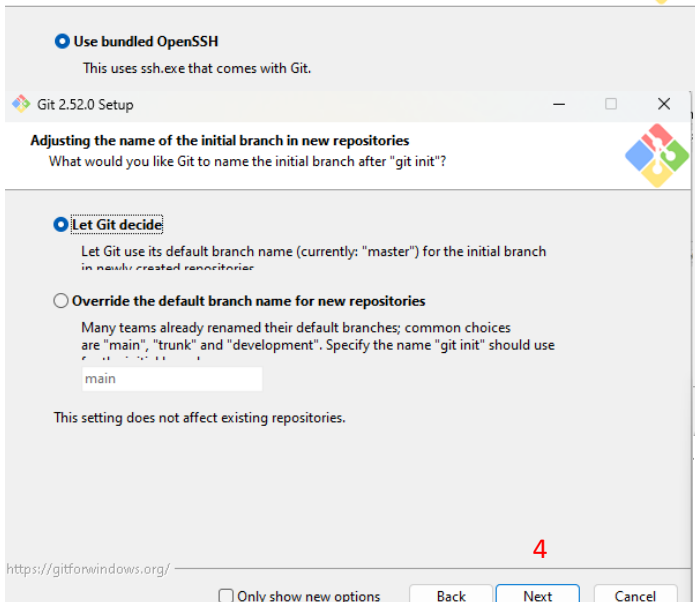
Check CPU architecture before downloading. If ARM, then select ARM64.

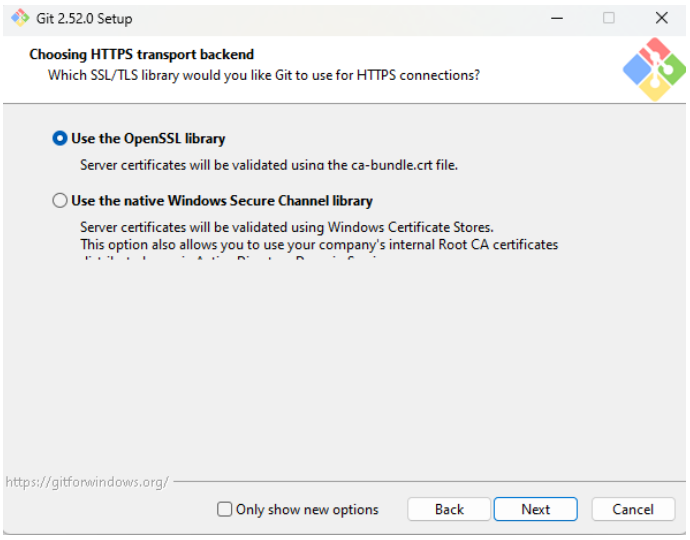
Run the installer and follow the prompts.

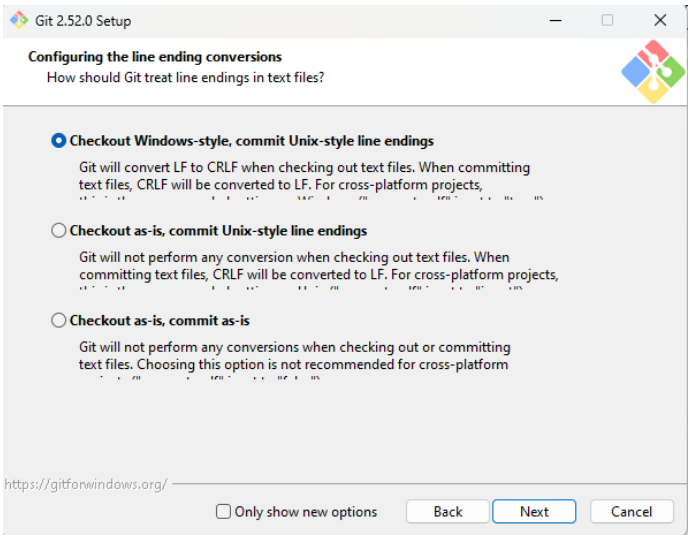


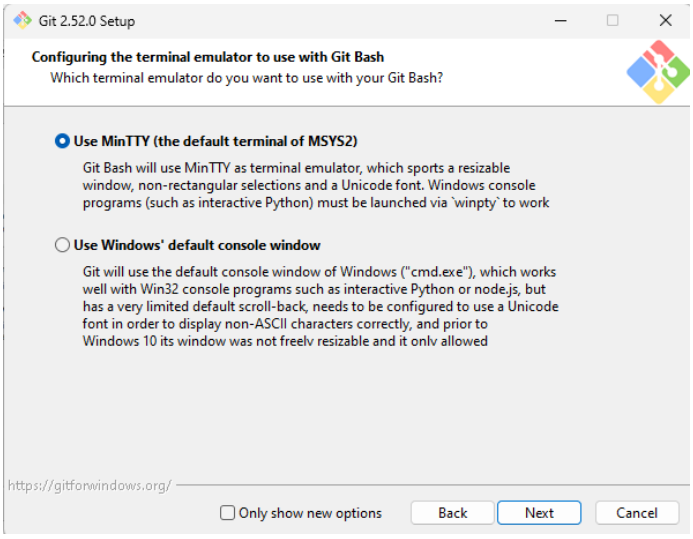


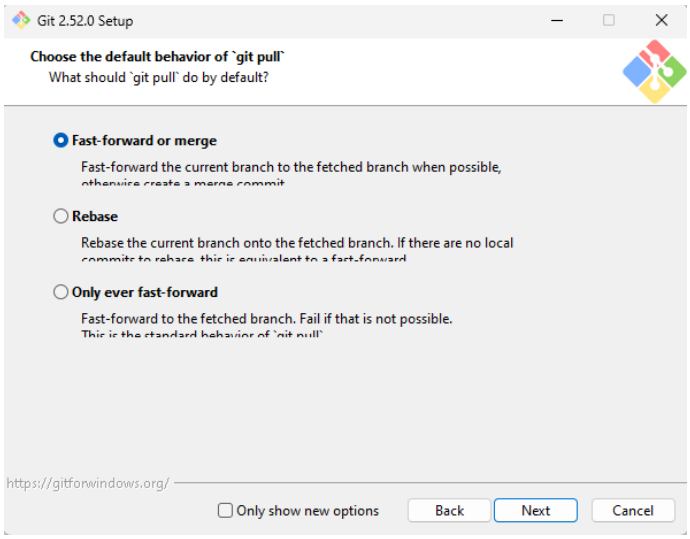


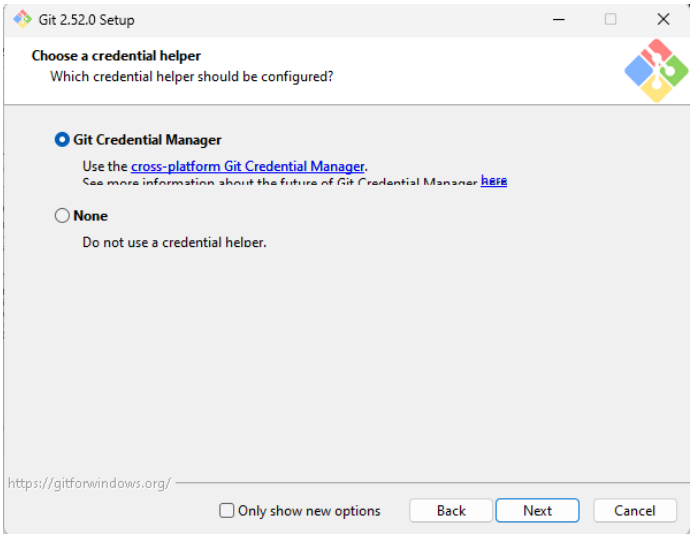


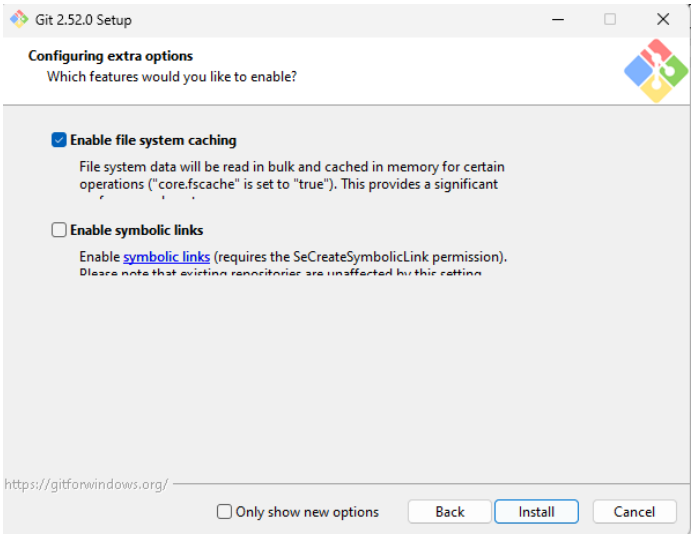

Git 2.52.0 Setup
Choosing HTTPS transport backend
Which SSL/TLS library would you like Git to use for HTTPS connections?
Use the OpenSSL library
Server certificates will be validated using the ca-bundle.crt file.
Use the native Windows Secure Channel library
Server certificates will be validated using Windows Certificate Stores.
This option also allows you to use your company's internal Root CA certificates
https://gitforwindows.org/
Only show new options
Back
Next
Cancel

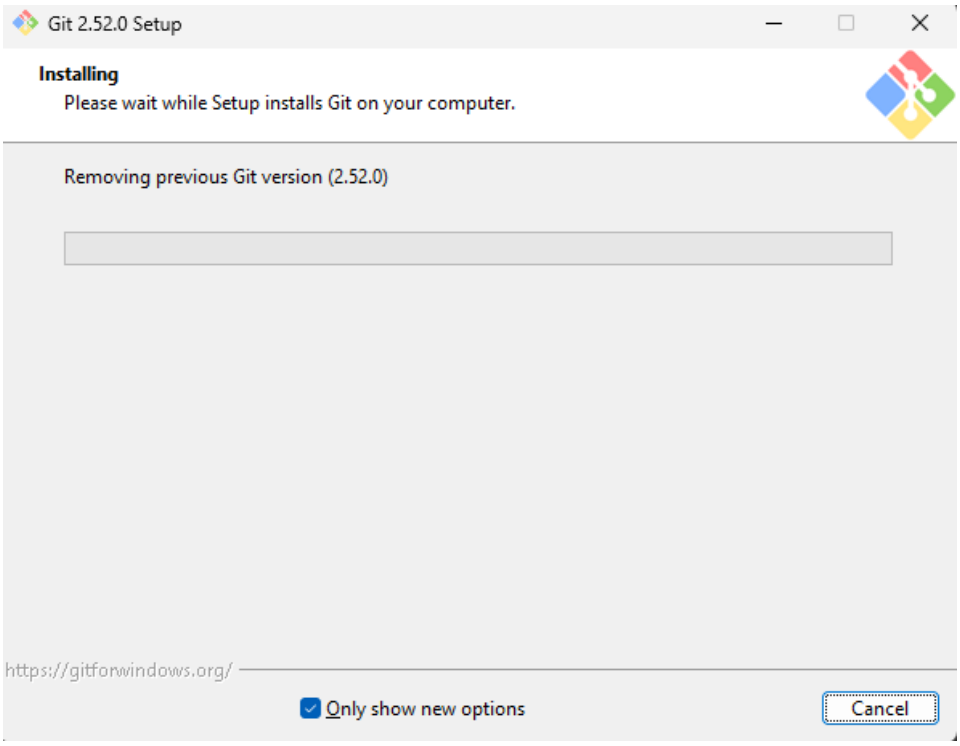

Git 2.52.0 Setup
Configuring the line ending conversions
How should Git treat line endings in text files?
Checkout Windows-style, commit Unix-style line endings
Git will convert LF to CRLF when checking out text files. When committing
text files, CRLF will be converted to LF. For cross-platform projects,
Checkout as-is, commit Unix-style line endings
Git will not perform any conversion when checking out text files. When
committing text files, CRLF will be converted to LF. For cross-platform projects,
Checkout as-is, commit as-is
Git will not perform any conversions when checking out or committing
text files. Choosing this option is not recommended for cross-platform
https://gitforwindows.org/
Only show new options
Back
Next
Cancel


Git 2.52.0 Setup
Configuring the terminal emulator to use with Git Bash
Which terminal emulator do you want to use with your Git Bash?
Use MinTTY (the default terminal of MSYS2)
Git Bash will use MinTTY as terminal emulator, which sports a resizable
window, non-rectangular selections and a Unicode font. Windows console
programs (such as interactive Python) must be launched via `winpty` to work
Use Windows' default console window
Git will use the default console window of Windows ("cmd.exe"), which works
well with Win32 console programs such as interactive Python or node.js, but
has a very limited default scroll-back, needs to be configured to use a Unicode
font in order to display non-ASCII characters correctly, and prior to
Windows 10 its window was not freely resizable and it only allowed
https://gitforwindows.org/
Only show new options
Back
Next
Cancel

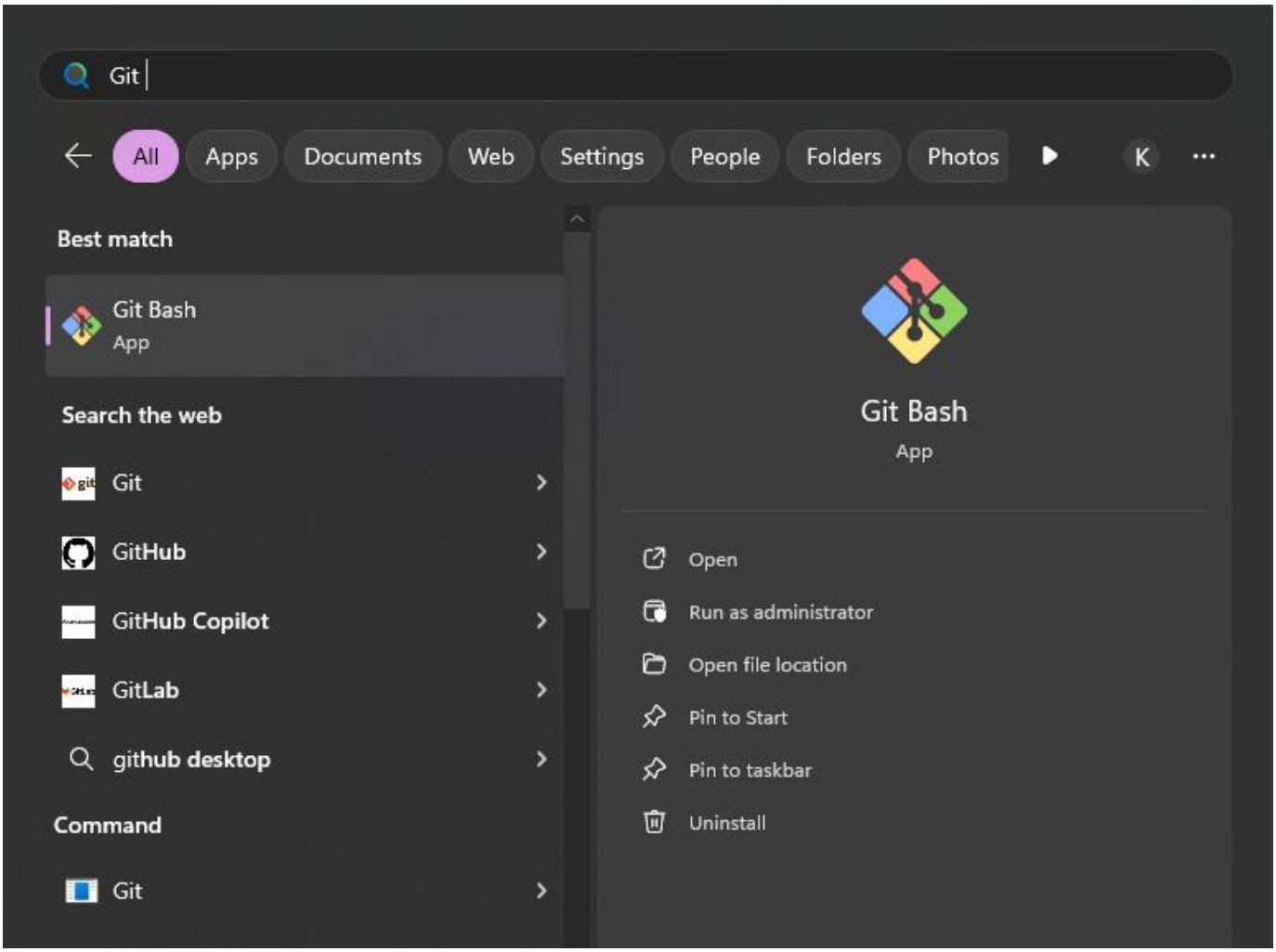

Git 2.52.0 Setup
Choose the default behavior of `git pull`
What should `git pull` do by default?
Fast-forward or merge
Fast-forward the current branch to the fetched branch when possible,
Rebase
Rebase the current branch onto the fetched branch. If there are no local
Only ever fast-forward
Fast-forward to the fetched branch. Fail if that is not possible.
https://gitforwindows.org/
Only show new options
Back
Next
Cancel


Git 2.52.0 Setup
Choose a credential helper
Which credential helper should be configured?
Git Credential Manager
Use the cross-platform Git Credential Manager.
None
Do not use a credential helper.
https://gitforwindows.org/
Only show new options
Back
Next
Cancel


Git 2.52.0 Setup
Configuring extra options
Which features would you like to enable?
Enable file system caching
File system data will be read in bulk and cached in memory for certain
operations ("core.fscache" is set to "true"). This provides a significant
Enable symbolic links
Enable symbolic links (requires the SeCreateSymbolicLink permission).
https://gitforwindows.org/
Only show new options
Back
Install
Cancel

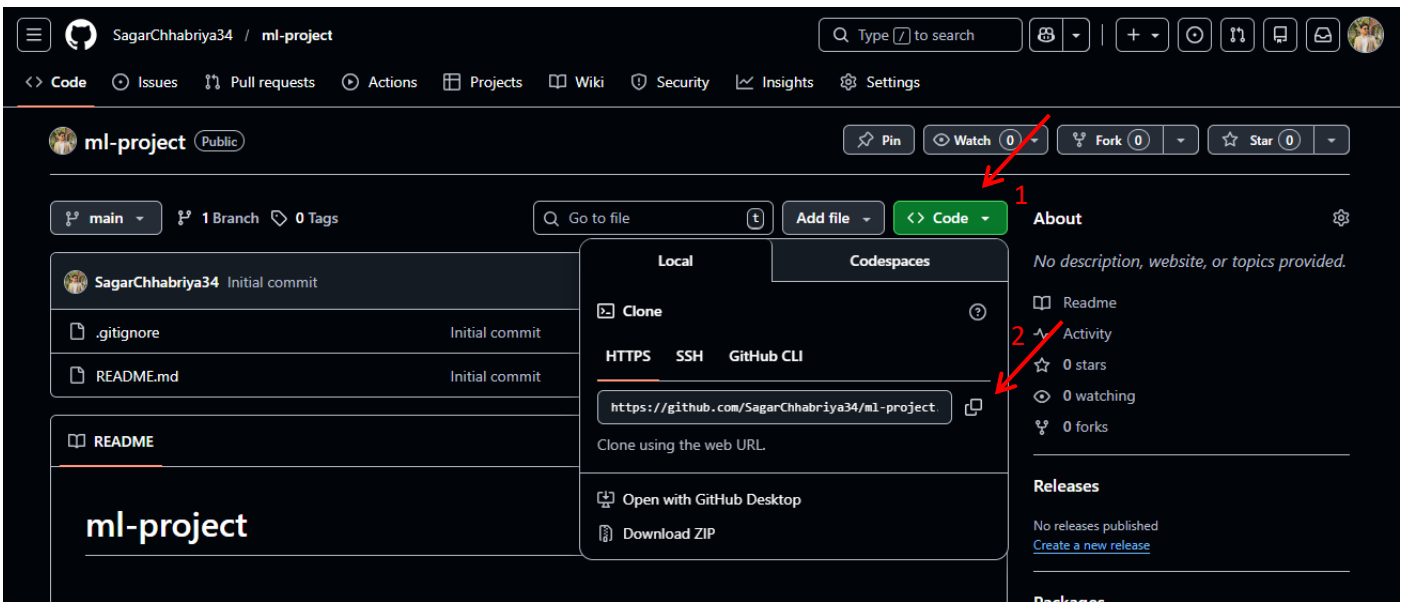

Git 2.52.0 Setup
Installing
Please wait while Setup installs Git on your computer.
Removing previous Git version (2.52.0)
https://gitforwindows.org/
Only show new options
Cancel

After installation, you can open Git Bash by searching for it in the Windows Start menu.



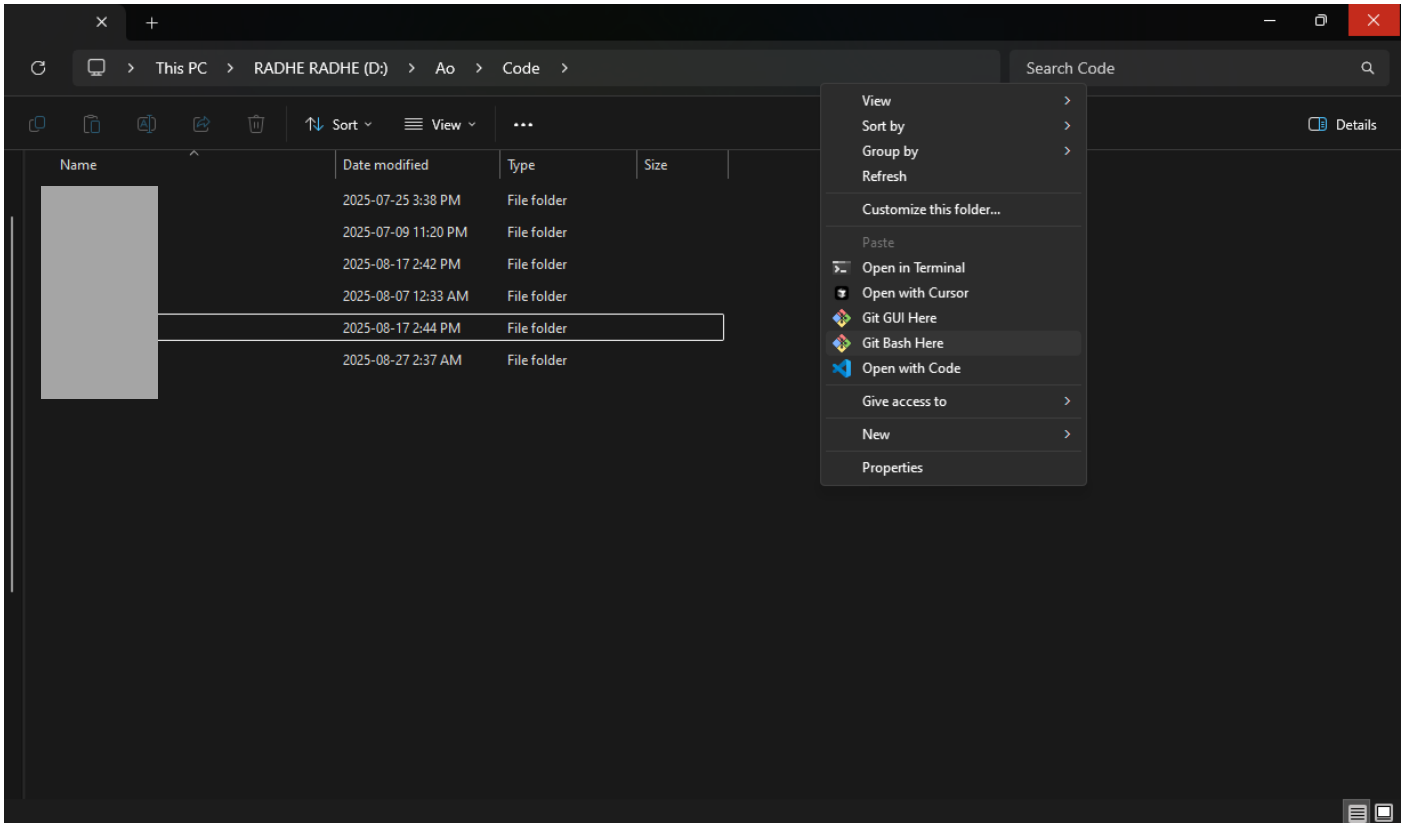
2.2 Clone the Repository

1. Go to the GitHub repository page that you created in Step 1.
2. Click the green **"Code"** button and copy the HTTPS clone link.

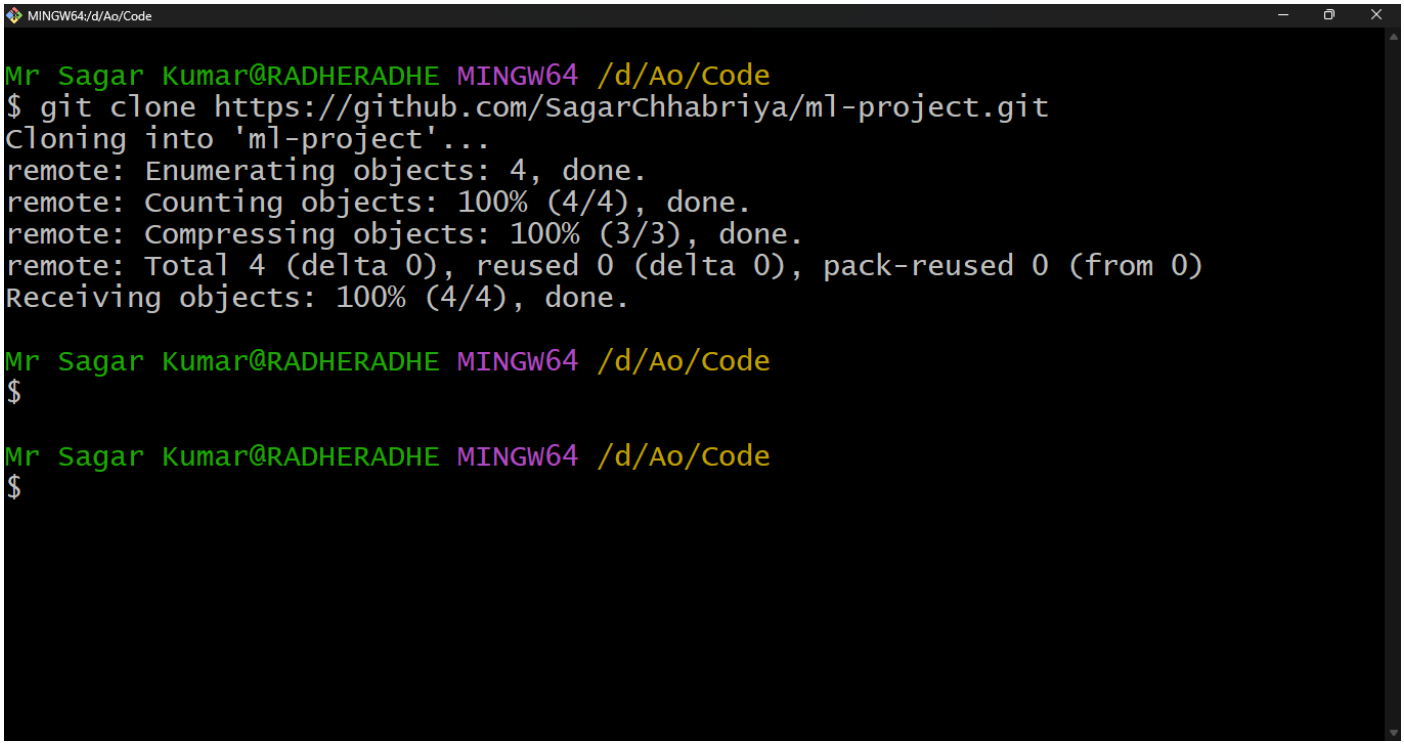


3. Open **Git Bash** (or your terminal of choice).

   Open any folder > Right Click > Git Bash Here



Type git clone <repo link > and hit enter.

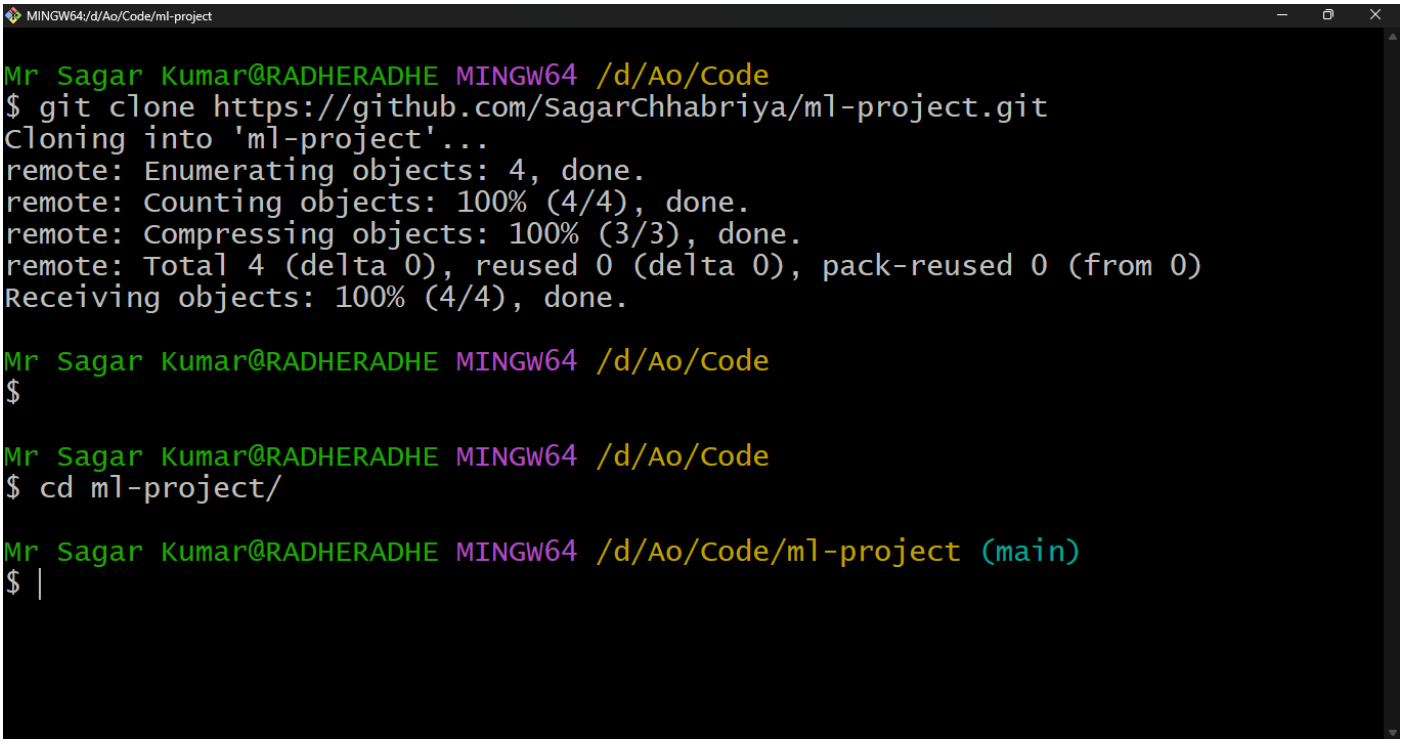


Note: You need to config the email and username with git bash for authorization after installing it first time. Run the following commands:

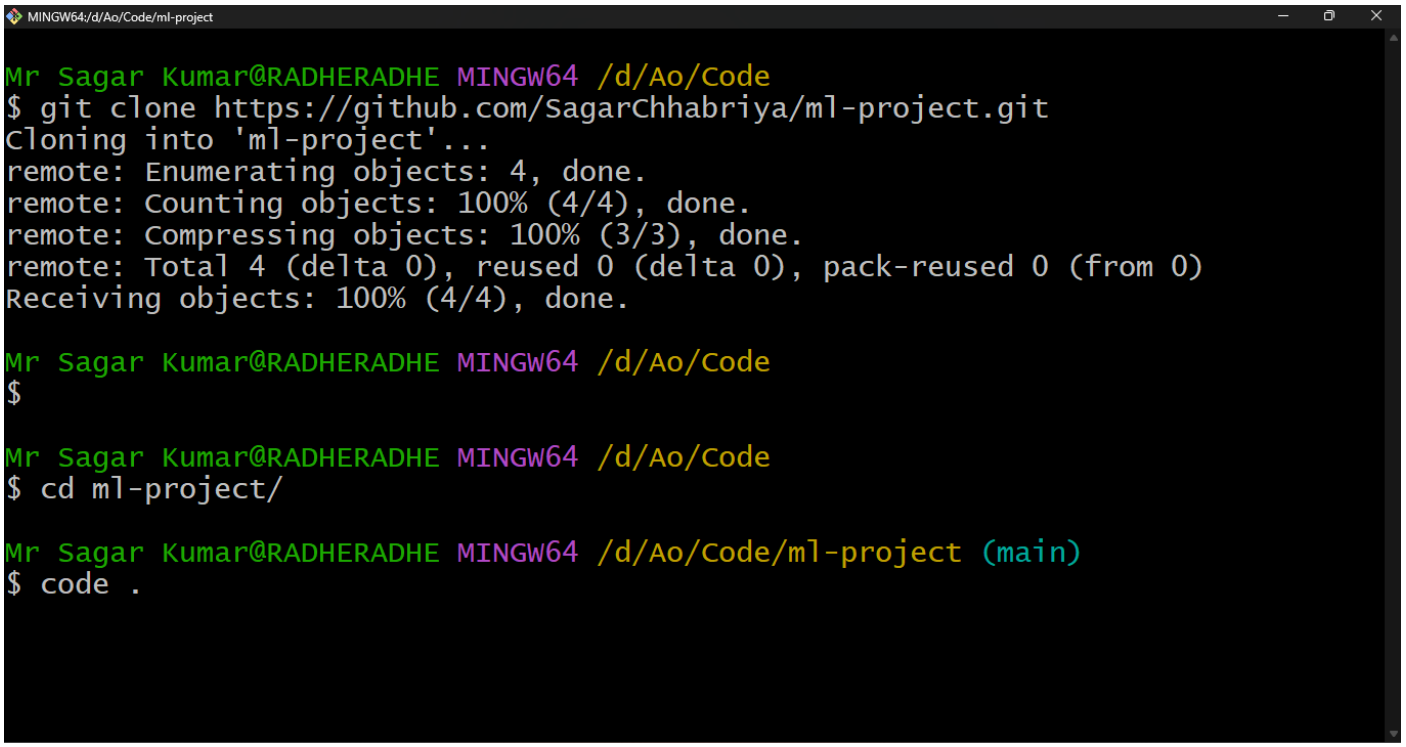
- git config --global user.email “youemail@org.com”
- git config --global user.name “yourGitHubUserHandle”

4. Navigate to the cloned repository directory using the terminal: cd ml-project

   Note: If you're not familiar with commands, move to the next page.

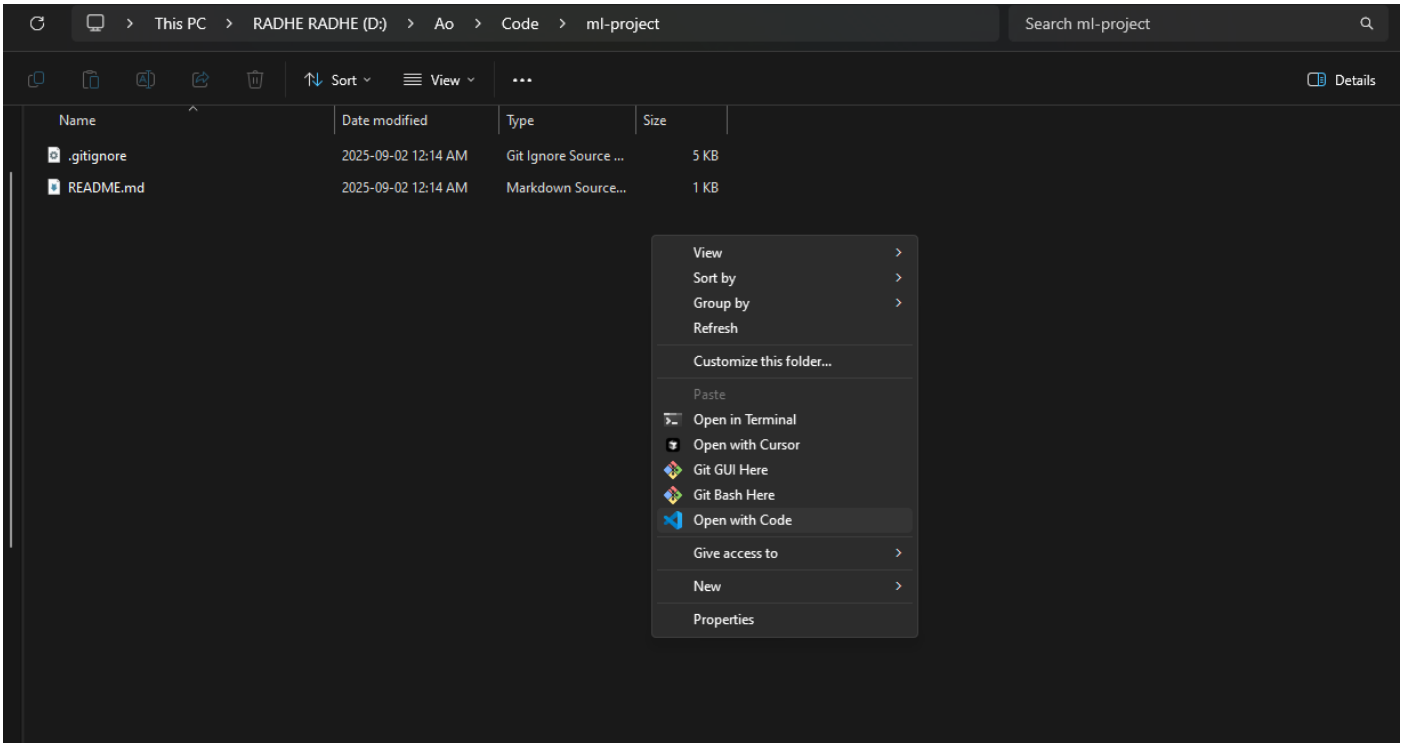


5. Open the repository in VS Code: type **code .** and hit enter. The VS Code will be opened auto.

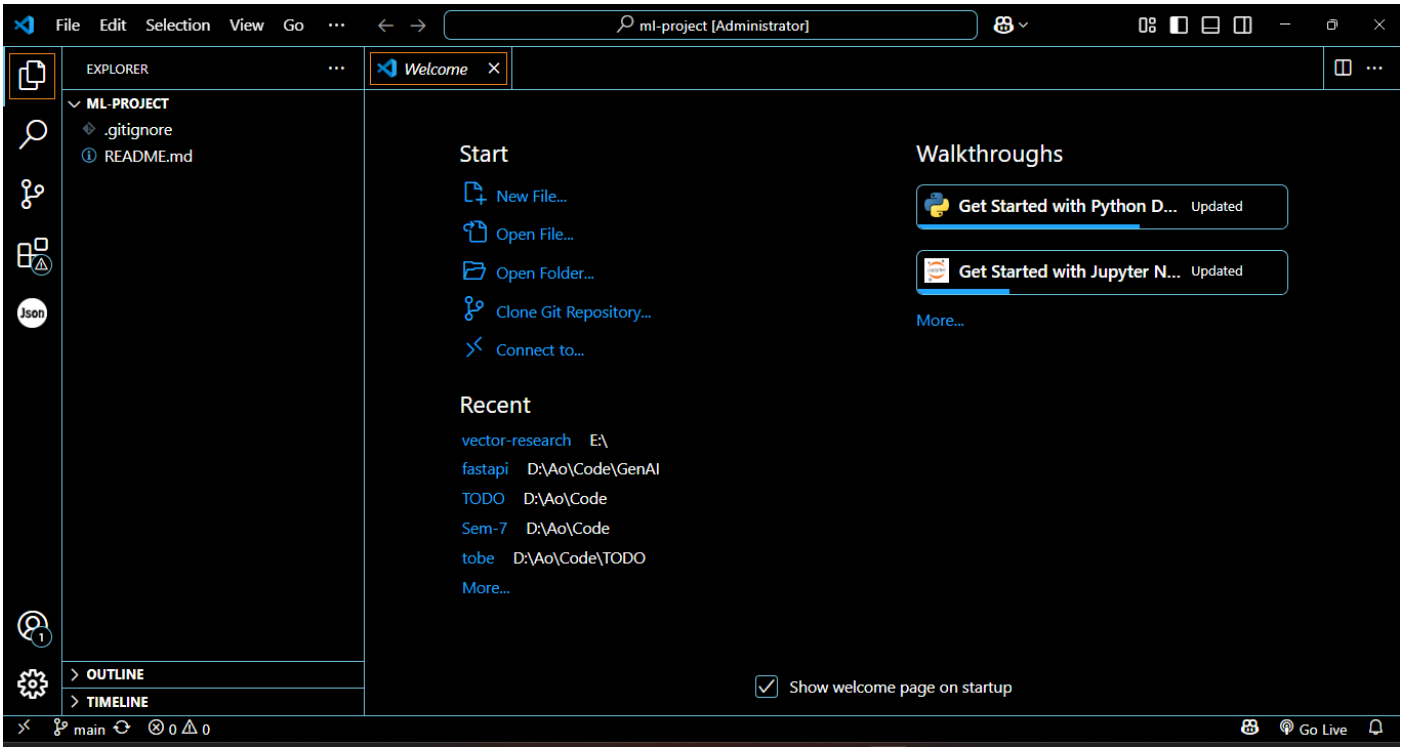


If you're not familiar with the commands, we also have an alternative to steps 4 and 5.

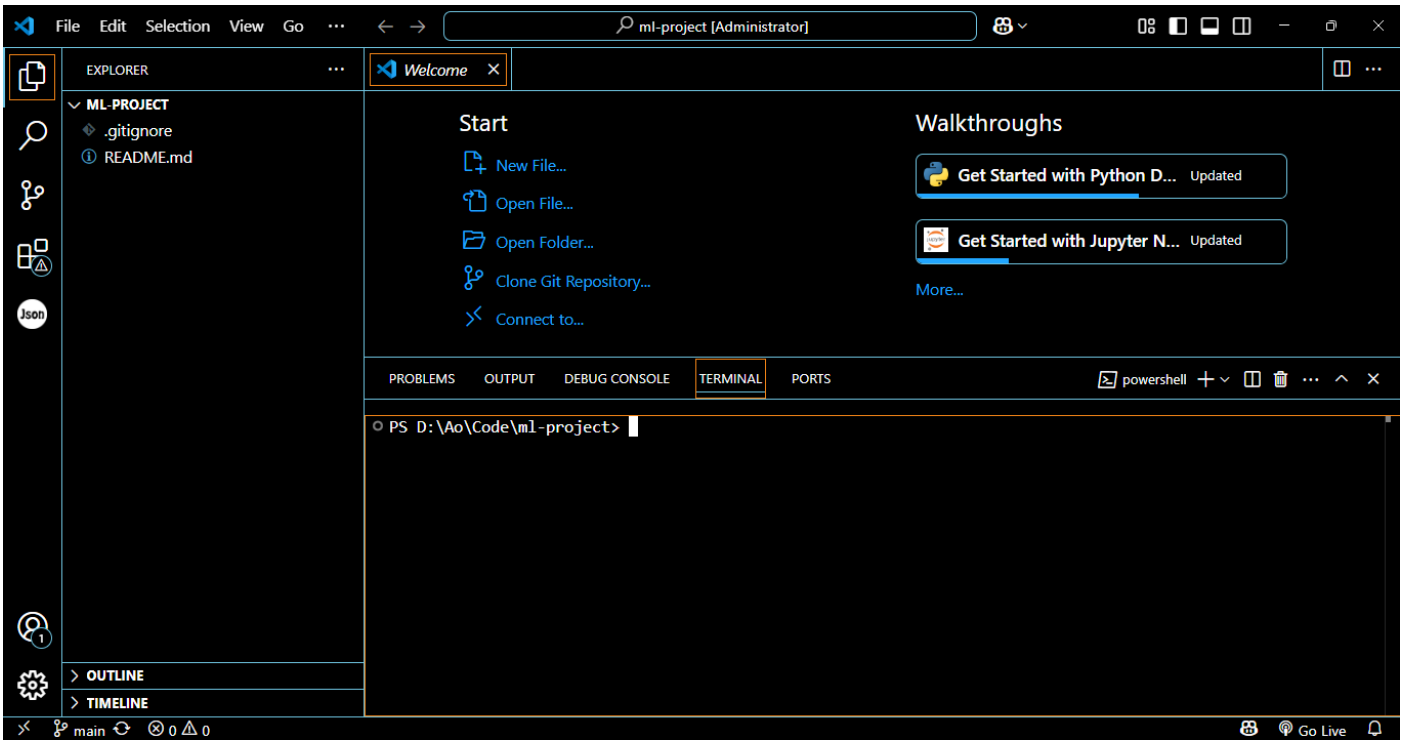
Open the project folder (i.e., directory) > Right Click > Open with Code



VS Code Preview after opening in the project directory.



6. In VS Code, open the integrated terminal by pressing Ctrl + J (or use the menu Terminal > New Terminal). And follow the next steps.



2.3 Create and Activate a Virtual Environment

1. **Create Virtual Environment**:

   To isolate project dependencies, create a virtual environment (venv) in your project directory:

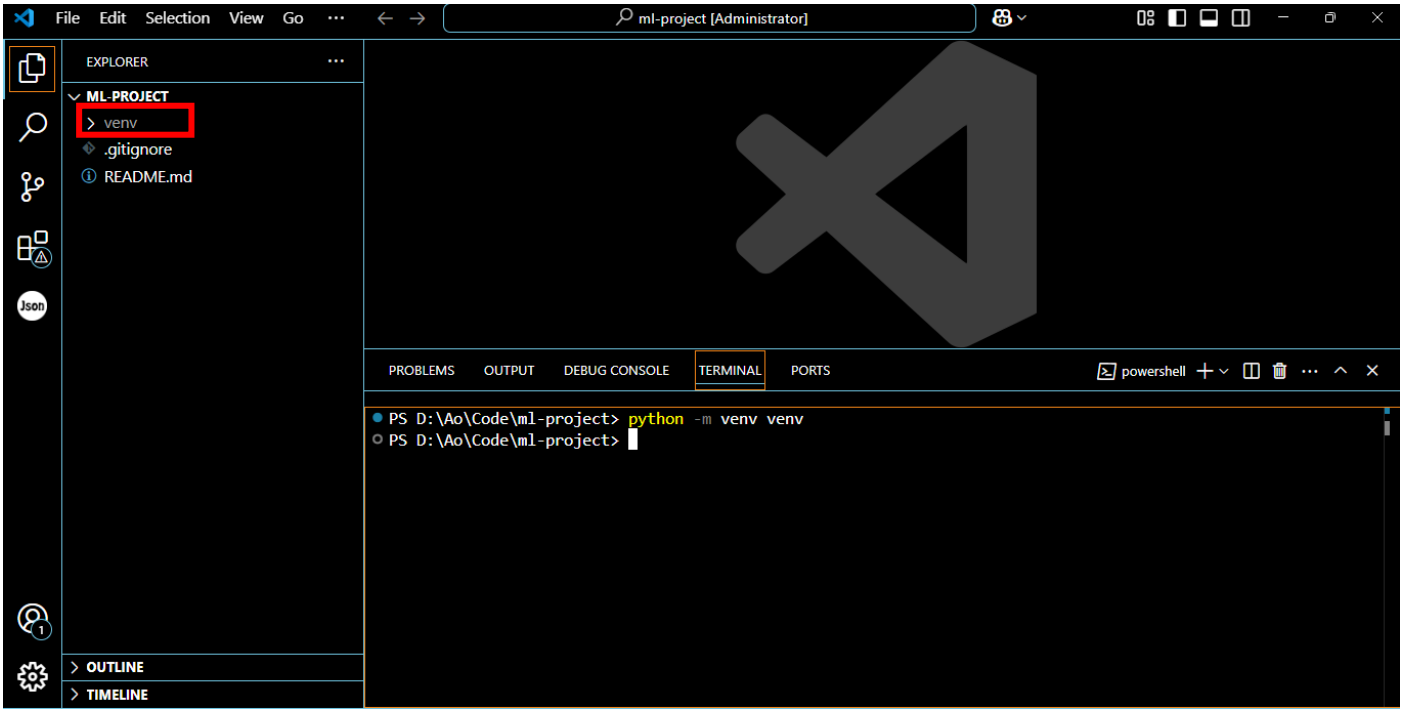
2. python -m venv my_venv

   Replace my_venv with the desired name for your virtual environment. Make sure python is installed and the also the required extensions.

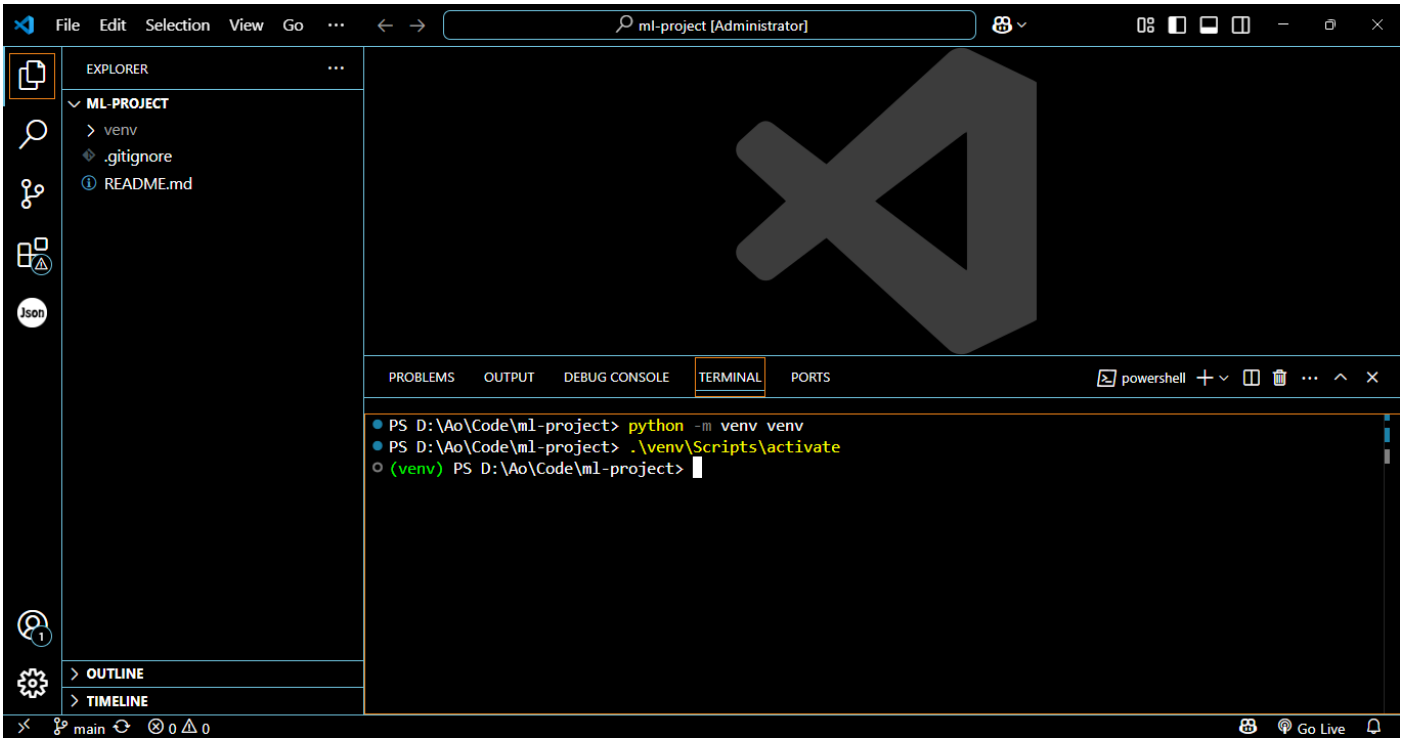
3. **Activate the Virtual Environment**:
    - On Windows:

      my_venv\Scripts\activate
    - On macOS/Linux:

      source my_venv/bin/activate



Environment Activated (indicated by the (venv))



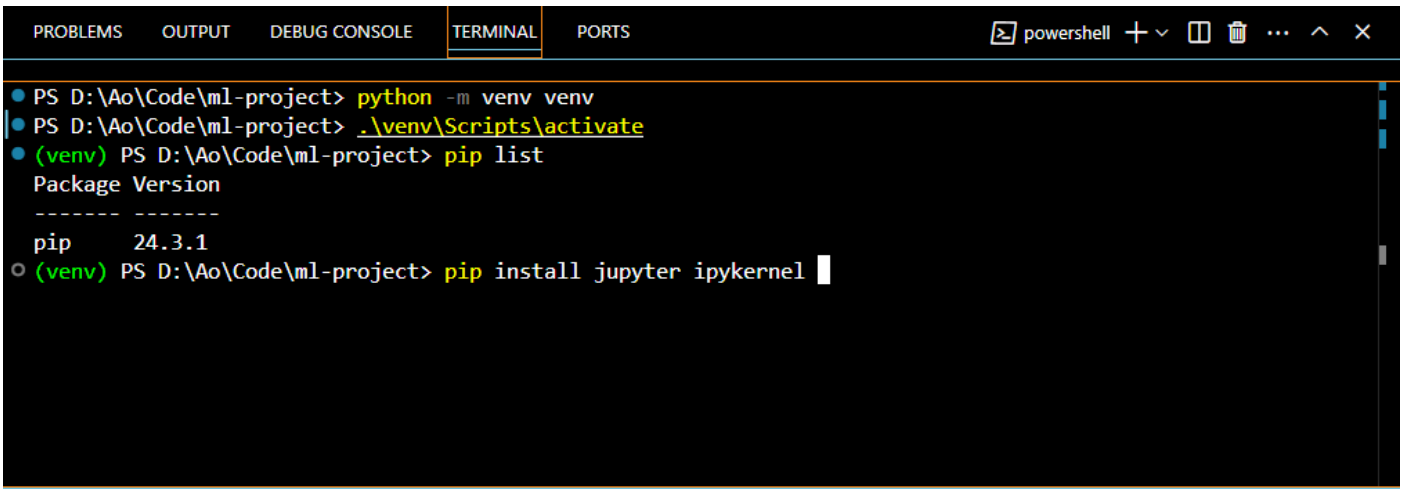
4. If you encounter the error **"not found"** in the PowerShell terminal of VS Code, follow these steps:
    - Run the following command to change the execution policy in PowerShell:
        - Set-ExecutionPolicy RemoteSigned -Scope Process
    - After running the above command, activate the virtual environment again:
        - my_venv\Scripts\activate

2.4 Install Required Python Libraries

Once the virtual environment is activated, you can install the required libraries for the project:

pip list *will display the installed packages. In this case only pip is listed that is the default package.*

1. Install the necessary libraries by running:
2. pip install jupyter ipykernel pandas numpy scikit-learn and hit enter



3. To install additional libraries, use the following command format:
    - For a single library:
        - pip install <library_name>
    - For multiple libraries in a batch:
        - pip install <library1> <library2> <library3> ...

This completes the setup of your local environment. You are now ready to start working on the project, using the virtual environment to manage project-specific dependencies and tools.

3. Model Selection, Training and Evaluation

Step 3. Model Selection, Training and Evaluation

Step 3 focuses on model selection, training, evaluation, and optimization using a Jupyter notebook to enable interactive experimentation and analysis. At this stage, all model development and experimentation will be performed inside the notebook for ease of understanding and rapid iteration. In later stages of the bootcamp, you will learn how to refactor this workflow into standalone Python scripts, separating experimentation from production-ready training code. This transition, from notebook-based experimentation to script-based model training, will be covered in detail in the MLOps module, where best practices for scalable, maintainable, and automated machine learning pipelines will be introduced.

Create a notebook file named model.ipynb and follow the instructions given below:

3.1 Load Data and Import Libraries

1. Import Libraries**:** Begin by importing the necessary libraries for data manipulation, visualization, and machine learning. Common libraries include pandas, numpy, matplotlib, seaborn, and scikit-learn, among others.

2. Load Data**:** Load the dataset into a pandas DataFrame. Ensure the data is loaded correctly and is in a format suitable for preprocessing.

3.2 Data Preprocessing, Exploratory Data Analysis (EDA), and Train-Test Split

1. Data Preprocessing: Clean the data by handling missing values, encoding categorical variables, and scaling numerical features. Apply transformations as needed based on the type of data.

2. Exploratory Data Analysis (EDA): Perform an initial analysis of the data to understand distributions, identify relationships, and spot outliers. This may involve generating summary statistics and visualizations. Handle the outliers and multicollinearity.

3. Train-Test Split: Split the dataset into training and testing sets. The typical ratio for splitting is 80/20 or 70/30, ensuring that the model can be trained on one portion and evaluated on another.

3.3 Model Training

1. Model Selection: Choose at least **three** machine learning models based on the task (e.g., classification, regression). Common models include Linear Regression, Logistic Regression, Decision Tree (or RF), Support Vector Machines, or KNN.

2. Train the Base Models: Fit each model using the default parameters (call them base models). Ensure that all the base models are initialized and trained on the available features and target variable.

3.4 Model Evaluation

1. Evaluate Performance: Assess the model's performance using appropriate evaluation metrics.

    - Classification: Consider metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

    - Regression: Evaluate using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), or R-squared.

If the task is classification then use the classification_report method of scikit-learn and store the results in a file named classification_report.json. For regression, do the same, store the results a file called model_performance.json

2. Perform additional validation:

    - K-Fold Cross-Validation

        - Split data into k fold (commonly 5 or 10)
        - Train on k-1 and validate on the remaining fold.
        - Compute and report the average score across all folds.

    - Stratified K-Fold Cross-Validation

        - Preserve class distribution in each fold.
        - Must be used for imbalanced classification datasets.

3.5 Testing on New Instances

After model training and evaluation, test the model on new data points (instances) in a Jupyter notebook environment. This can be done by passing individual or small batches of new data through the model to generate predictions interactively.

3.6 Model Optimization: Hyperparameter Tuning

Improve models' performance by experimenting with hyperparameter optimization techniques such as grid search or random search (depends on the data and the machine). Use cross validation during tuning. Test the tuned models on new data point (i.e., test dataset) and demonstrate inference inside the jupyter notebook.

3.7 Model Serialization

Once the model has been trained and optimized, serialize it using a standard format (e.g., Pickle or Joblib). This step allows the model to be saved to a file, making it available for later use without the need for retraining. Name the file algo_best_model.pkl such as linreg_best_model.pkl.

Now load the saved model into memory and make predictions, ensuring that the same version of the model is used consistently.

4. Streamlit UI Development

Step 4. Streamlit UI Development (app.py)

The user interface for the model will be built using **Streamlit**, which allows for quick and easy creation of interactive web applications for data science projects. The following tasks outline the steps to set up the Streamlit UI: Create a file named streamlit_app.py

4.1 Import Libraries

1. Begin by importing the required libraries for building the app, including:

    - Streamlit for UI components and interaction.
    - Libraries like joblib or pickle for loading the trained model.

    Example libraries to import:

    - streamlit as st
    - joblib (or pickle for model loading)
    - Any other necessary libraries for handling data preprocessing or visualization.

4.2 Set Page Title and Document Title

1. Page Title: Use Streamlit's built-in functions to set the title of the app's main page.

2. Document Title: Optionally, set the document title for the browser tab, which will appear in the title bar of the user's browser.

4.3 Load Model

Load the serialized model that was saved earlier (in Step 3) using a library like joblib or pickle. This allows the Streamlit app to use the model for making predictions.

4.4 Get Input via Streamlit Input Functions

Use Streamlit's input functions to gather user input for making predictions. This may include:

- Text Input: For entering numeric or text-based data.
- Selectbox or Radio Buttons: For selecting categorical data.
- Sliders: For selecting numerical values within a range.

4.5 Make Predictions and Display Results

- Pass the collected user input to the loaded model and make predictions based on the input data.

- After obtaining the prediction, display the results to the user using Streamlit's built-in functions like st.success(), st.error(), or st.write(). These functions provide simple ways to communicate the results back to the user in a clear and informative way.

5. Prepare Model for Deployment

Step 5. Deployment

Once the model and Streamlit application are developed, the next step is to prepare the project for deployment. This section outlines the steps for creating the requirements.txt file and ensuring the correct dependencies for deployment.

5.1 Define requirements.txt

The requirements.txt file lists all the necessary Python libraries and their corresponding versions that are required to run the application and make predictions.

Identify the version of each library used in the project, such as numpy, pandas, scikit-learn, streamlit, and any other dependencies required for the app. This can be done by importing each package in a Python script or notebook and calling the module.__version__ attribute.

Example:

- import numpy as np
- print(np.__version__) # Returns the version of numpy

Based on the version information obtained, manually create the requirements.txt file by specifying each library and its version. For example:

```
numpy==1.21.2
pandas==1.3.3
scikit-learn==0.24.2
streamlit==0.87.0
joblib==1.0.1
```

Note: Avoid using `pip freeze > requirements.txt`, as it may include unnecessary libraries that were not used during the project. And make sure the filename is correct *requirement**s.**txt.*

5.2 Prepare for Deployment

Verify that all the libraries you have used throughout the project, including those required for model loading, data preprocessing, and the Streamlit UI, are included in the requirements.txt file.

Finalize project structure: Ensure that the project folder contains all necessary files (such as app.py, model.pkl, and requirements.txt) in an organized manner, ready for deployment.

6. Deploy on Streamlit Community Cloud

Step 6. Streamlit Community Cloud

The Streamlit Community Cloud provides a platform to deploy your Streamlit app online for public access. This section outlines how to deploy the app on Streamlit Community Cloud using GitHub.

6.1 Continue with GitHub

1. **Push Code to GitHub**:

   Before deploying on Streamlit, ensure your project code (including app.py, requirements.txt, and any other necessary files) is pushed to the GitHub repository (ml-project). This is the very repository we created in the first step. https://graphite.com/guides/how-to-push-code-from-vscode-to-github

2. **Ensure Public Access**:

   Make sure the repository is public or that you have set the correct access permissions to allow Streamlit Community Cloud to access it.

6.2 Create Streamlit App

1. **Sign in to Streamlit Community Cloud**:
   Go to Streamlit Community Cloud and sign in using your GitHub account. Select the Free.

2. **Create a New App**:

   - Click on **"Create app"** from the top right
   - Select: Deploy a public app from GitHub
   - Select the GitHub repository where your project is stored.
   - Choose the main file for your Streamlit app (e.g., app.py).
   - Set the App URL
   - Set up any other configuration settings as needed such as python version.

3. **Deploy the App**:

   Once the repository is linked, Streamlit will automatically deploy your app. You will be provided with a unique URL where your app can be accessed.

4. **Test the Deployment**:

   After deployment, test the app on the provided URL to ensure that the model works correctly, the UI is responsive, and the predictions are accurate.

End of Project